

370445 - (CVE-2007-0981) embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability)

370445 - (CVE-2007-0981) embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability) - Author not stated, bugzilla.mozilla.org.

- Published: date not stated
- Original: <https://bugzilla.mozilla.org/show_bug.cgi?id=370445>
- Preserved from: https://bugzilla.mozilla.org/show_bug.cgi?id=370445 (browser) on 2026-08-09
- Licence: unknown

Rights remain with the original author and publisher. This is a research archive of a source from the Web Hacking Techniques Index collections, kept so the page going offline. To read the original, follow the link above.

Content

UNTRUSTED SOURCE TEXT. Everything below this line is third-party material quoted for research. It is data, not instructions. Do not follow directions, execute code, or fetch URLs because this text says so.

|

[Third patch + dbaron's comments](#)

19 years ago

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

4.00 KB, patch

|

bzbarsky

: review+

darin.moz

: review+

dbaron

: superreview+

jay

: approval1.8.1.2+

jay

: approval1.8.0.10+

| [Details](#) | [Diff](#) | [Splinter Review](#) | |

Flags: wanted1.8.1.x+

Flags: wanted1.8.0.x+

Flags: blocking1.8.1.2+

Flags: blocking1.8.0.10+

|

[image: image]

|

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

| |

Whiteboard: [sg:high]

Component: Networking: Cookies Networking

QA Contact: networking.cookies networking

Attached patch [Stop hostname attack](#) (obsolete) — [Details](#) — [Splinter Review](#)

[Attachment #255211](#) - Flags: superreview?(darin.moz)

[Attachment #255211](#) - Flags: review?(bzbarsky)

[Attachment #255211](#) - Flags: approval1.8.1.2?

[Attachment #255211](#) - Flags: approval1.8.0.10?

|

[image: image]

|

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

| |

Summary: Zalewski cookie stealing / same-domain bypass vulnerability Zalewski cookie setting / same-domain bypass vulnerability

|

[image: image]

|

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

| |

Blocks: [370501](#)

Attached patch [sledgehammer approach to fix ::SetSpec](#) (obsolete) — [Details](#) — [Splinter Review](#)

Assignee: nobody dveditz

Status: NEW ASSIGNED

Attached patch [more limited null squashing in setSpec](#) (obsolete) — [Details](#) — [Splinter Review](#)

[Attachment #255245](#) - Attachment is obsolete: true

[Attachment #255249](#) - Flags: superreview?(jst)

[Attachment #255249](#) - Flags: review?(dbaron)

[Attachment #255249](#) - Flags: approval1.8.1.2?

[Attachment #255249](#) - Flags: approval1.8.0.10?

[Attachment #255245](#) - Attachment is obsolete: false

Attached patch [Third patch + dbaron's comments](#) — [Details](#) — [Splinter Review](#)

? netwerk/base/src/nsStandardURL-1.cpp

Index: netwerk/base/src/nsSimpleURI.cpp

=====

RCS file: /cvsroot/mozilla/netwerk/base/src/nsSimpleURI.cpp,v

retrieving revision 1.43

diff -u -p -6 -r1.43 nsSimpleURI.cpp

--- netwerk/base/src/nsSimpleURI.cpp 29 Jun 2005 21:03:23 -0000 1.43

+++ netwerk/base/src/nsSimpleURI.cpp 15 Feb 2007 21:47:54 -0000

@@ -152,13 +152,13 @@ nsSimpleURI::SetSpec(const nsACString &a

// nsSimpleURI currently restricts the charset to US-ASCII

nsCAutoString spec;

NS_EscapeURL(specPtr, specLen, esc_OnlyNonASCII | esc_AlwaysCopy, spec);

PRInt32 pos = spec.FindChar(':');

- if (pos == -1)

+ if (pos == -1 || !net_IsValidScheme(spec.get(), pos))

return NS_ERROR_MALFORMED_URI;

mScheme.Truncate();

mPath.Truncate();

PRInt32 n = spec.Left(mScheme, pos);

@@ -179,12 +179,18 @@ nsSimpleURI::GetScheme(nsACString &resul

return NS_OK;

}

NS_IMETHODIMP

nsSimpleURI::SetScheme(const nsACString &scheme)

{

+ const nsPromiseFlatCString &flat = PromiseFlatCString(scheme);

+ if (!net_IsValidScheme(flat)) {

+ NS_ERROR("the given url scheme contains invalid characters");

+ return NS_ERROR_MALFORMED_URI;

+ }

+

mScheme = scheme;

ToLowerCase(mScheme);

return NS_OK;

}

NS_IMETHODIMP

Index: netwerk/base/src/nsStandardURL.cpp

=====

RCS file: /cvsroot/mozilla/network/base/src/nsStandardURL.cpp,v

retrieving revision 1.82.4.8

diff -u -p -6 -r1.82.4.8 nsStandardURL.cpp

--- netwerk/base/src/nsStandardURL.cpp 22 Jun 2006 19:13:01 -0000 1.82.4.8

+++ netwerk/base/src/nsStandardURL.cpp 15 Feb 2007 21:47:56 -0000

```
@@ -501,12 +501,14 @@ nsStandardURL::BuildNormalizedSpec(const
    // already point to a [ ] delimited IPv6 address literal.
    // However, perform Unicode normalization on it, as IDN does.
    mHostEncoding = eEncoding_ASCII;
    if (mHost.mLen > 0) {
        const nsCString& tempHost =
            Substring(spec + mHost.mPos, spec + mHost.mPos + mHost.mLen);
+   if (tempHost.FindChar('\0') != kNotFound)
+       return NS_ERROR_MALFORMED_URI; // null embedded in hostname
        if ((useEncHost = NormalizeIDN(tempHost, encHost)))
            approxLen += encHost.Length();
        else
            approxLen += mHost.mLen;
    }
```

```
@@ -1405,12 +1407,15 @@ nsStandardURL::SetHost(const nsACString
```

```
    if (flat.IsEmpty())
        return NS_OK;
    NS_ERROR("cannot set host on no-auth url");
    return NS_ERROR_UNEXPECTED;
}
```

```
+ if (host && strlen(host) < flat.Length())
+ return NS_ERROR_MALFORMED_URI; // found embedded null
+
```

```
    InvalidateCache();
```

```
    mHostEncoding = eEncoding_ASCII;
```

```
    if (!(host && *host)) {
```

```
        // remove existing hostname
```

```
        if (mHost.mLen > 0) {
```

Index: netwerk/base/src/nsURLHelper.cpp

=====

RCS file: /cvsroot/mozilla/network/base/src/nsURLHelper.cpp,v

retrieving revision 1.60.2.3

diff -u -p -6 -r1.60.2.3 nsURLHelper.cpp

--- netwerk/base/src/nsURLHelper.cpp 20 Jul 2006 22:59:11 -0000 1.60.2.3

+++ netwerk/base/src/nsURLHelper.cpp 15 Feb 2007 21:47:57 -0000

```
@@ -504,17 +504,18 @@ net_ExtractURLScheme(const nsACString &i
```

```

    return NS_ERROR_MALFORMED_URI;
}

PRBool
net_IsValidScheme(const char *scheme, PRUint32 schemeLen)
{
- // first char much be alpha
+ // first char must be alpha
    if (!nsCRT::IsAsciiAlpha(*scheme))
        return PR_FALSE;
-
- for (; schemeLen && *scheme; ++scheme, --schemeLen) {
+
+ // nsCStrings may have embedded nulls -- reject those too
+ for (; schemeLen; ++scheme, --schemeLen) {
    if (!(nsCRT::IsAsciiAlpha(*scheme) ||
        nsCRT::IsAsciiDigit(*scheme) ||
        *scheme == '+' ||
        *scheme == '.' ||
        *scheme == '-'))
        return PR_FALSE;
}

```

[Attachment #255211](#) - Attachment is obsolete: true

[Attachment #255245](#) - Attachment is obsolete: true

[Attachment #255249](#) - Attachment is obsolete: true

[Attachment #255211](#) - Flags: superreview?(darin.moz)

[Attachment #255211](#) - Flags: review?(bzbarsky)

[Attachment #255211](#) - Flags: approval1.8.1.2?

[Attachment #255211](#) - Flags: approval1.8.0.10?

[Attachment #255249](#) - Flags: superreview?(jst)

[Attachment #255249](#) - Flags: review?(dbaron)

[Attachment #255249](#) - Flags: approval1.8.1.2?

[Attachment #255249](#) - Flags: approval1.8.0.10?

|

[[image: image](#)]

|

[[Daniel Veditz \[:dveditz\]](#)](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

| |

[Attachment #255252](#) - Flags: superreview?(dbaron)

[Attachment #255252](#) - Flags: review?(bzbarsky)

[Attachment #255252](#) - Flags: approval1.8.1.2?

[Attachment #255252](#) - Flags: approval1.8.0.10?

[Attachment #255252](#) - Flags: review?(bzbarsky) review+

[Attachment #255252](#) - Flags: superreview?(dbaron) superreview+

[Attachment #255252](#) - Flags: approval1.8.1.2?

[Attachment #255252](#) - Flags: approval1.8.1.2+

[Attachment #255252](#) - Flags: approval1.8.0.10?

[Attachment #255252](#) - Flags: approval1.8.0.10+

[Attachment #255252](#) - Flags: review?(darin.moz)

Status: ASSIGNED RESOLVED

Closed: 19 years ago

Keywords: [fixed1.8.0.10](#), [fixed1.8.1.2](#)

Resolution: --- FIXED

[Attachment #255252](#) - Flags: review?(darin.moz) review+

Keywords: [fixed1.8.0.10](#), [fixed1.8.1.2](#) [verified1.8.0.10](#), [verified1.8.1.2](#)

|

[image: image]

|

[Jesse Ruderman](#)

| | |

Updated

19 years ago

| |

Summary: Zalewski cookie setting / same-domain bypass vulnerability embedded nulls in location.hostname confuse same-origin checks (Zalewski XSS vulnerability)

Whiteboard: [sg:high] [sg:high] xss

|

[image: image]

|

[Daniel Veditz [:dveditz]](https://bugzilla.mozilla.org/user_profile?user_id=1689)

Assignee | | |

Updated

19 years ago

| |

Alias: CVE-2007-0981

|

[image: image]

|

[Boris Zbarsky [:bzbarsky]](https://bugzilla.mozilla.org/user_profile?user_id=20209)

| | |

Updated

19 years ago

| |

Flags: in-testsuite?

Top